THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The sum of 5 consecutive integers is 180. The largest of them is: 1

(a) 39 **(b)** 37 **(c)** 40 **(d)** 38

2. The average monthly salary of 22 employees is ₹66 thousand. An employee earning ₹25 thousand leaves and a new employee joins, and the average rises by ₹2 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 71 **(b)** 69 **(c)** 74 **(d)** 67

3. Set r = 0. While n > 0: r = 10r + (n mod 10), then n = quotient of n by 10. Starting with n = 5702, the final r is: 1

(a) 2165 **(b)** 2075 **(c)** 2066 **(d)** 2084

4. A rhombus has side 51 cm and one diagonal 48 cm. The other diagonal is: 1

(a) 88 cm **(b)** 90 cm **(c)** 92 cm **(d)** 135 cm

5. The points $(-5, 19)$, $(-4, 18)$ and $(-3, k)$ are collinear. The value of k is: 1

(a) 16 **(b)** 17 **(c)** 18 **(d)** 19

6. A cuboid 54 cm $\times$ 12 cm $\times$ 6 cm is melted and recast into cubes of edge 3 cm. The number of cubes is: 1

(a) 146 **(b)** 148 **(c)** 144 **(d)** 150

7. If the distance between $(x, 2)$ and $(1, 6)$ is 5 and $x > 1$, then $x =$ 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 6

8. If $x = 4$ is the zero of $kx - 28$, then $k =$ 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

9. $1.\overline{27}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{14}{11}$ **(b)** $\frac{13}{11}$ **(c)** $\frac{7}{6}$ **(d)** $\frac{15}{11}$

10. If $x + y = 10$ and $x^2 + y^2 = 58$, then $xy =$ 1

(a) 22 **(b)** 20 **(c)** 21 **(d)** 23

11. The exterior angle at D of cyclic quadrilateral $ABCD$ (side CD produced) equals the interior angle at B, which is $72°$. The interior angle $\angle ADC$ is: 1

(a) $108°$ **(b)** $128°$ **(c)** $118°$ **(d)** $98°$

12. The area of a rhombus of side 13 cm and one diagonal 10 cm is (cm^2): 1

(a) 121 **(b)** 122 **(c)** 120 **(d)** 119

13. A bag has 36 balls, red or blue. If $P(\text{red}) = \frac{5}{12}$, the number of blue balls is: 1

(a) 20 **(b)** 23 **(c)** 22 **(d)** 21

14. Three numbers in AP have sum 30 and product 840 (with $d > 0$). The largest number is: 1

(a) 15 **(b)** 10 **(c)** 18 **(d)** 14

15. If n is any integer not divisible by 3, then n^2 leaves a remainder ___ on division by 3. For 7 this gives: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 0

16. The average of 48 numbers is 75. The average of all 100 numbers (including 52 more numbers) is 88. The average of the 52 additional numbers is: 1

(a) 101 **(b)** 100 **(c)** 99 **(d)** 102

17. Using $\gcd(82, 298) = 2$, the LCM of 82 and 298 is: 1

(a) 12216 **(b)** 12220 **(c)** 12218 **(d)** 12300

18. The diagonals of a rhombus are 28 cm and 96 cm. Its area is: 1

(a) 1372 cm^2 **(b)** 1440 cm^2 **(c)** 1344 cm^2 **(d)** 1354 cm^2

19. The line $kx + 2y = 16$ passes through $(1, 4)$. Its slope is: 1

(a) -2 **(b)** -3 **(c)** -4 **(d)** -5

20. A cylinder of radius 7 cm and height 15 cm is melted and recast into cones of the same radius and height 1 cm. The number of cones is: 1

(a) 44 **(b)** 46 **(c)** 45 **(d)** 60

21. The fourth vertex of the parallelogram with consecutive vertices $(1, 1)$, $(4, 2)$, $(6, 5)$ (in order) is: 1

(a) $(3, 4)$ **(b)** $(-3, 4)$ **(c)** $(3, -4)$ **(d)** $(4, 3)$

22. If a linear polynomial $ax + b$ has zero 0, then: 1

(a) $a = b$ **(b)** $b = 0$ **(c)** $b = 1$ **(d)** $a = 0$

23. The value of $0.\overline{9}$ is: 1

(a) 0.99 **(b)** just less than 1

(c) 1 **(d)** 0.9

24. If $x - y = 3$ and $x^2 + y^2 = 65$, then $xy =$ 1

(a) 27 **(b)** 28 **(c)** 30 **(d)** 29

25. AB is a diameter, C and D are on the circle (same side of AB) and $ABCD$ is cyclic with $\angle DAB = 48^\circ$. Then $\angle BCD =$ 1

(a) 122° **(b)** 132° **(c)** 152° **(d)** 142°

26. A triangle has sides 10 cm, 17 cm, 21 cm. The length of the shortest altitude (cm) is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 7

27. A letter is chosen from the word MISSISSIPPI. The probability that it is not the letter I is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{8}{11}$ **(c)** $\frac{7}{12}$ **(d)** $\frac{7}{11}$

28. A town's population is 10000 and grows by 10% each year. After 2 years it is: 1

(a) 12100 **(b)** 12000 **(c)** 12300 **(d)** 12200

29. For the two-digit number 83, the value of (number − digit sum) ÷ 9 is: 1

(a) 10 **(b)** 9 **(c)** 7 **(d)** 8

30. A shop earns a profit of 13% on sales of ₹2000 and 24% on sales of ₹9000. The overall profit percentage on total sales is: 1

(a) 22% **(b)** 23% **(c)** 24% **(d)** 21%

31. To test whether 2381 is prime, we check divisibility by 2, 3, 4, … up to $\lfloor\sqrt{2381}\rfloor$. The last number checked is: 1

(a) 1190 **(b)** 49 **(c)** 47 **(d)** 48

32. The diagonal of a rectangle is 25 cm and one side is 7 cm. Its area is: 1

(a) 168 cm^2 **(b)** 192 cm^2 **(c)** 178 cm^2 **(d)** 175 cm^2

33. The line $9x + 5y = 45$ has intercepts p on the x-axis and q on the y-axis. The value of $p + q$ is: 1

(a) 59 **(b)** 13 **(c)** 14 **(d)** 15

34. Take $\pi = \frac{22}{7}$. The capacity of a hemispherical bowl of inner radius 10.5 cm is: 1

(a) 2668.5 cm^3 **(b)** 2912.5 cm^3 **(c)** 2182.5 cm^3 **(d)** 2425.5 cm^3

35. A point $P(x, 0)$ on the x-axis is 10 units from $(-3, 8)$ with $x > -3$. The value of x is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

36. For the pattern with nth term $6n - 1$, the sum of the 1st and 10th terms is: 1

(a) 63 **(b)** 64 **(c)** 65 **(d)** 66

37. The length of the repeating block of $\frac{1}{41}$ is: 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

38. If $x^2 + \frac{1}{x^2} = 14$, then $\left(x + \frac{1}{x}\right)^2 =$ 1

(a) 16 **(b)** 15 **(c)** 17 **(d)** 18

39. A chord AB of length 14 cm lies 24 cm from the centre O. The area of triangle OAB (cm^2) is: 1

(a) 167 **(b)** 168 **(c)** 169 **(d)** 170

40. A rope of length 88 cm (take $\pi = \frac{22}{7}$) forms a circle. The same rope forms a square. The area of the square (cm^2) is: 1

(a) 486 **(b)** 484 **(c)** 483 **(d)** 485

41. **Assertion (A):** The greatest common divisor of 48 and 18 is 6. 1

Reason (R): The gcd of two numbers is the largest number that divides both of them.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** The sum of the angles of a quadrilateral is 360°. 1

Reason (R): A diagonal divides a quadrilateral into two triangles, each with angle sum 180°.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** The pair $(2, 3)$ is a solution of $3x + 2y = 12$. 1

Reason (R): A pair (x, y) is a solution of an equation if substituting it makes both sides equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** The volume of a cube of side 5 cm is 125 cm^3. 1

Reason (R): The volume of a cube of side a is a^3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** The point $(0, -6)$ lies on the y-axis. 1
Reason (R): Every point on the x-axis has ordinate 0.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The graph of $y = 2x - 6$ cuts the x-axis at $(3, 0)$. 1
Reason (R): The graph of $y = 2x - 6$ cuts the y-axis at $(0, -6)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The number $\sqrt{2}$ is irrational. 1
Reason (R): The decimal expansion of an irrational number is non-terminating and non-repeating.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** $x^2 - 9 = (x - 3)(x + 3)$. 1
Reason (R): $(a + b)^2 = a^2 + 2ab + b^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** Equal chords of a circle are equidistant from the centre. 1
Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The circumference of a circle of radius 7 cm is 44 cm (take $\pi = \frac{22}{7}$). 1
Reason (R): The area of a circle of radius r is πr^2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**